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Monte Carlo Portfolio Risk — Methodology

Mathematical framework, model assumptions, and validation approach

Monte Carlo Portfolio Risk Simulator — Methodology

Objective

Estimate a distribution of hypothetical terminal portfolio values and downside risk measures using Monte Carlo simulation calibrated to historical returns.

Portfolio

Five assets are held at equal weights: AAPL, AMZN, GOOGL, MSFT, and NVDA. Weights are reset to equal weights every simulated trading day. The S&P 500 benchmark is not included as a holding in this simulation.

Data and return transformations

Given adjusted or prepared prices (P_{i,t}), calculate log returns:

[r_{i,t} = \ln(P_{i,t}/P_{i,t-1})]

Estimate the sample mean vector (\hat{\mu}) and covariance matrix (\hat{\Sigma}) from the observed daily log-return series.

Monte Carlo model

For every simulation and day, generate correlated asset log returns:

$$ \mathbf{r}_t \sim \mathcal{N}(\hat{\mu},\hat{\Sigma}) $$

Convert them to simple returns:

$$ R_{i,t} = e^{r_{i,t}}-1 $$

Calculate daily equal-weight portfolio returns and compound wealth:

$$ R_{p,t} = \sum_i w_i R_{i,t}, \qquad V_T = V_0 \prod_{t=1}^{T}(1+R_{p,t}) $$

The covariance matrix is symmetrized and its eigenvalues are floored at a small positive value for numerical stability before sampling.

Risk metrics

For terminal simple return ($R_T=V_T/V_0-1$), at confidence level (c):

$$ q_{1-c}=Q_{1-c}(R_T) $$

The signed VaR loss amount is ($-V_0 q_{1-c}$). Expected Shortfall is estimated as the negative initial-capital-scaled mean of terminal returns at or below the lower-tail quantile.

Probability of loss is the proportion of simulations with ($V_T<V_0$). Terminal value percentiles summarize the modeled distribution, not guaranteed future bounds.

Important distinctions

- The model simulates a joint distribution of returns; it does not predict a specific market path.
- Correlation is estimated from the historical covariance matrix, but may not remain stable during stress.
- The Gaussian assumption can understate fat tails and extreme joint events.
- Historical mean and covariance estimates are uncertain.
- No transaction costs, taxes, slippage, or liquidity constraints are included.
- The output is not investment advice or a validated production risk model.

Suggested validation

1. Compare simulated and historical marginal return distributions.
2. Compare model VaR with realized out-of-sample losses.
3. Perform VaR exception counting and independence/conditional coverage tests.
4. Test block bootstrap and Student-t alternatives.
5. Add volatility models and explicit stress scenarios.
6. Compare results under different estimation windows and rebalancing schedules.